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Science for resilient livelihoods in dry areas



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Planning and Designing Microcatchment Rainwater Harvesting Systems for Agriculture

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WEBINARS SERIES ON RAINWATER HARVESTING

26 July 2022

Module n°4: Session 2: Planning and
Designing of Rainwater Harvesting
Systems and Structures



1/2022

Recovering lost rainwater: The concept of rainwater harvesting

No intervention



Moderate intervention



Appropriate intervention



Water Harvesting system components

1. The catchment

- Micro
- Macro

4. The storage

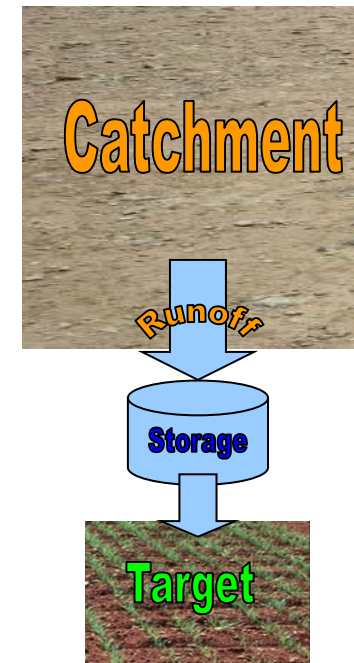
- Surface
- Soil
- Ground

2. The runoff

- Natural
- Induced

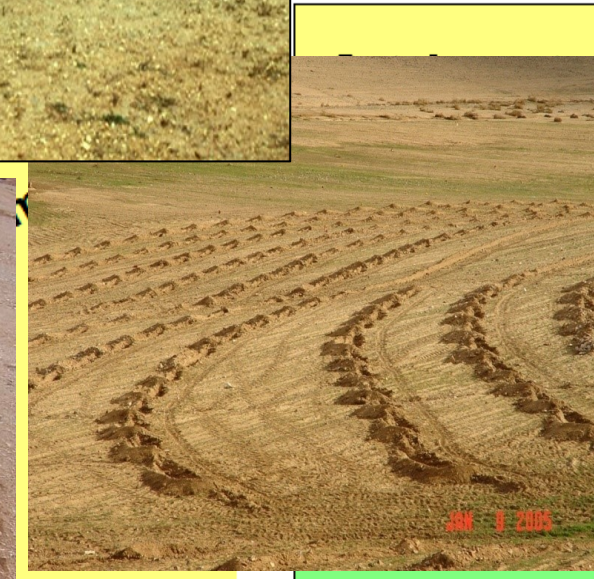
3. The target

- Agriculture
- domestic
- industrial
- environment



What is the design of micro-catchments rainwater harvesting systems?

- Mainly determining the size of the catchment relative to the size of the cropped area
- Also insure that the soil water storage can hold the harvested water



Micro-catchment rainwater harvesting: planning, design, and implementation: Training manual

ESCWA, Beirut, Lebanon

https://www.unescwa.org/sites/default/files/pubs/pdf/micro-catchment-rainwater-harvesting-training-manual-arabic_0.pdf

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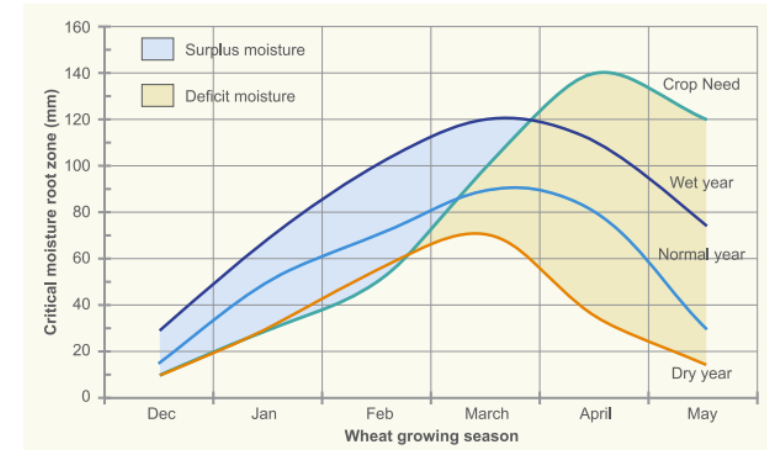
THEIB OWEIS



Step 1:

Determine the need and the potential for a water harvesting system

1. The need for a system arise when direct annual rainfall amounts and distribution are not enough to satisfy minimum crop water needs during the growing season in the absence of irrigation water
2. The potential for a system is governed by runoff possibility and catchment availability
3. When the rainwater harvesting project creates no conflict of interests (i.e. Upstream-downstream issues)
4. When institutions/people with enough capacity exist and willing to operate and maintain the system



Step 2:

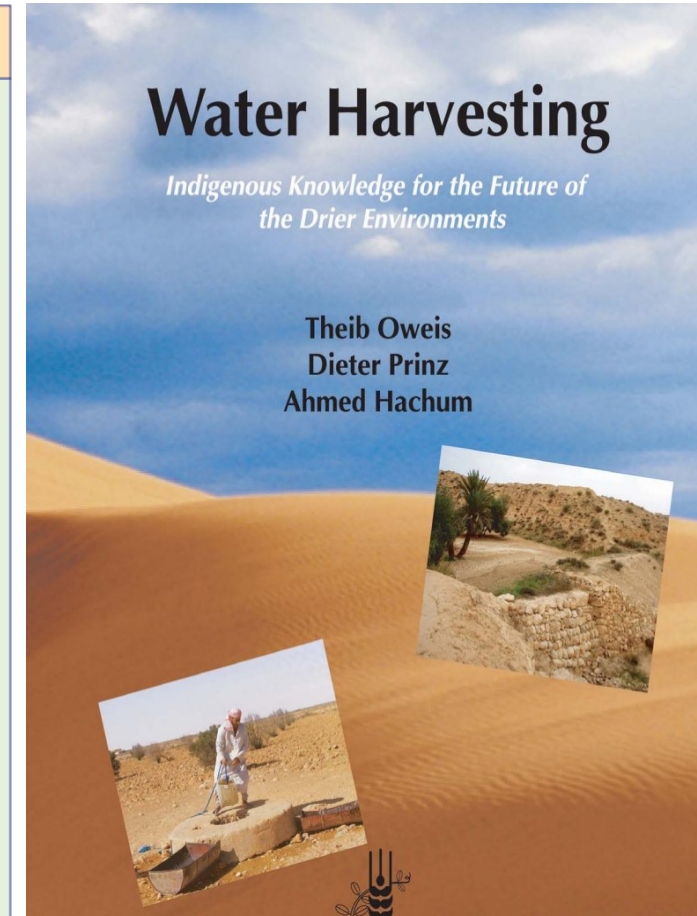
Select the technique suitable for the site and the crop

Micro catchment techniques	Common crops	Preferred land slope %	Preferred soil depth (cm)	Preferred soil type
Contour ridges and bunds	Shrubs and grasses, field crops,	1-25	≥ 50	Loamy sand to loam
Semicircular bunds and Negarim	Trees, shrubs	1-15	≥ 100	Loam to clay loam
Runoff strips	Field crops, grasses	0.5-10	≥60	Sandy loam to clay
Contour bench terrasses	Trees, shrubs and field crops	25-65	≥100	Loamy sand to loam
Rooftop systems	Domestic, livestock, home gardens	0-100	Not relevant	Not relevant

Guidelines for selecting water harvesting techniques in the dry areas

Technique	Crop	Soil		Land Slope (2)	Land Vegetation	Cover Stoniness (4)	Socio-economics				Storage type
		Depth (1)	Texture				Farm size (5)	Capital (6)	Labor (7)	Skill	
Micro-catchments											
On-farm systems											
Contour ridges	range	variable	variable	med.,steep	poor,med.	low-med.	variable	low	medium	local/	soil profile
	field	med.,deep	"	medium	poor	low	small,med.	"	"	training	"
	trees	deep	med.,heavy	low,med.	poor,med.	"	"	"	"	"	"
	vegetable	med.,deep	"	"	"	"	small	"	"	"	"
Semi-circular bunds (trapezoidal triangular)	range	"	variable	"	poor	low,med.	variable	"	high	local/no	"
	field	"	med.,heavy	"	"	"	small,med.	"	"	training	"
	trees	deep	"	"	poor,med.	low	"	"	"	"	"
	vegetable	"	"	"	"	"	small	"	"	"	"
Small pits	field	"	"	"	poor	"	variable	"	medium	"	"
	range	shallow,med.	"	"	poor,med.	low,med.	"	"	"	"	"
Small basins (Negarim)	range	med.,deep	"	"	poor	"	small	"	high	"	"
	trees	deep	"	low	poor,med.	"	small,med.	"	"	"	"
Runoff strips	range	variable	"	low,med.	poor	"	"	"	low	"	"
	field	med.,deep	"	"	poor,med.	"	"	"	"	"	"
Inter- row system (roaded catchment)	trees	deep	"	Low	poor	low	large	high	medium	local/training	"
	vegetable	medium	variable	"	"	"	"	"	"	"	"
	field	"	"	"	"	"	med.,large	"	"	"	"
Meskat (Khushkaba)	trees	deep	med., heavy	low,med.	poor,med.	"	variable	low	"	local/no	"
	field	medium	"	"	poor	"	small,med.	"	low	training	"
Contour bench terraces	trees	deep	"	Steep	"	"	"	high	medium	external skill	"
	field	medium	"	"	"	"	"	"	"	"	"
Rooftops	drinking	na	na	na	na	na	small	"	medium	local/training	surface/
	vegetable	variable	variable	"	"	"	"	"	"	"	subsurface
Macro-catchments											
Wadi-bed systems											
Small farm reservoirs	all crops	variable	"	low,med.	variable	variable	med.,large	high	high	external skill	surface/ subsurface
Wadi-bed cultivation	trees/ vegetable	med.,deep	med., heavy	"	poor	Low	small,med.	medium	med.,high	local	surface/soil profile
Jessour	trees	"	"	med.,steep	variable	variable	small	"	high	local/training	"
Off-wadi systems											
Water spreading	field/ trees	"	"	low,med.	poor	low,med.	variable	"	medium	external skill	soil profile
Large bunds	trees	deep	"	"	poor,med.	"	med.,large	"	"	local/training	"
	field	medium	"	"	poor	low	medium	"	"	"	"
	range	shallow,med.	variable	"	med.,dense	variable	med.,large	"	"	"	"
Tanks and hafair	all crops	variable	med., heavy	low	variable	"	"	med.,high	"	external skill	surface/subsurface
Cisterns	drinking/ trees/ vegetable	deep	rock	all slopes	poor,med.	variable	small,med.	medium	high	local/training	subsurface
Hillside runoff systems	field/ trees	med.,deep	med.,heavy	low,med.	poor,med.	low	small,med.	"	high	local/training	soil profile
(1) shallow < 50 cm, medium: 50-100 cm, deep> 100 cm; (2) low < 4%, medium: 4-12%, steep > 12%; (3) poor < 15%, medium: 15-30%, dense > 30% (4) low < 10%, medium: 10-25%, high> 25%; (5) small<5 ha, medium:5-25 ha, large> 25 ha; (6) low < \$ 25/ ha, medium: \$ 25-100/ ha, high > \$ 100/ ha; (7) low < 5 man-day/ ha, medium: 5-20 man-day/ ha, high > 20 man-day/ ha; na : not applicable.											

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Oweis, T., D. Prinz, and A. Hachum. 2001. *Water harvesting: Indigenous knowledge for the future of the drier environments*. ICARDA, Aleppo, Syria. 40pp.

Step 3:

Determine annual crop water requirements (Evapotranspiration; ET)

1. Direct method
2. Indirect methods
3. Secondary data
4. Averaged estimates

Averaged estimates of crop water requirements in MENA Mediterranean region

	Cereals	legumes	Trees	Grasses	Shrubs	Vegetables
Jan	65	45	20		10	
Feb	125	100	25	10	10	30
March	155	150	35	40	20	65
April	75	75	45	110	55	115
May	-	-	50	115	45	110
June	-	-	65	20	45	-
July	-	-	65	-	35	-
August	-	-	60	-	30	-
September	-	-	55	-	13	-
October	-	-	35	-	12	-
November	-	-	25	-	8	-
December	50	45	20	-	7	-
Total	470	415	500	295	290	320

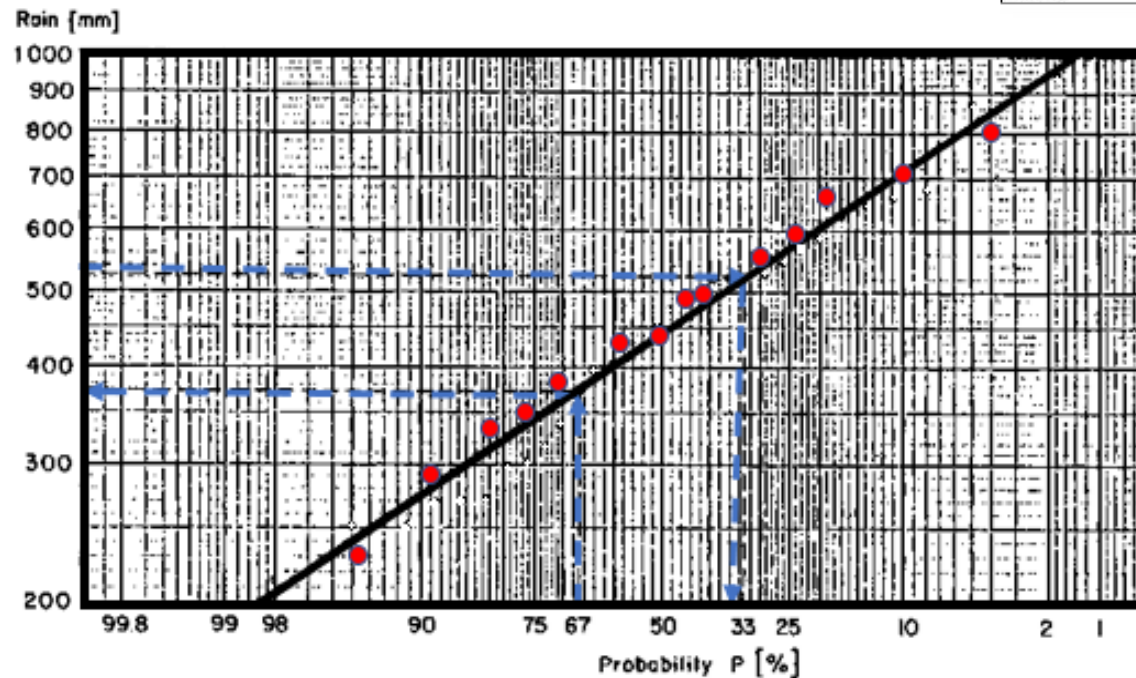
Step 4:

Determine design annual rainfall (R)

$$P(\%) = (m - 0.375) / (N + 0.25) * 100$$

Where: P = the probability in % of the observation of the rank (m) and N= the total number of observations.

Year	2020	2016	2011	2006	2009	2015	2013	2008	2017	2010	2012	2018	2019	2014	2007
Annual Rain	798	702	669	592	551	499	472	446	425	410	390	352	331	283	225
Rank (m)	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Probability (P%)	4.1	10.7	17.2	23.8	30.3	36.9	43.4	50.0	56.7	63.1	69.7	76.2	82.8	89.3	95.9



Station	No of years (N)	Mean annual rainfall (mm)	Probability 25%	probability 50%	Probability 67%	Probability 75%
Mwaqar	61	154	193	146	129	116
Sahab	51	254	294	235	203	193
Yaduda	47	313	369	300	262	253
Wadi shueib	59	355	432	346	279	254
Na'ur	72	442	571	409	369	326
Ajlun	77	632	749	641	558	500
Hummar	57	509	587	482	398	378

Step 5.

Determine design water storage capacity

Design storage capacity (m³) = Roots depth (m) X Roots area (m²) X Soil water holding capacity (WHC m/m)

Crop	Effective average root depth (m)	Effective average roots diameter (m)
Trees	1.2	1.0-4.0
shrubs	0.8	0.5-1.5
Cereals	1.0	Continuous
legumes	0.6	Continuous
vegetables	0.7	Rows of 1.0 m wide
grasses	0.5	Continuous

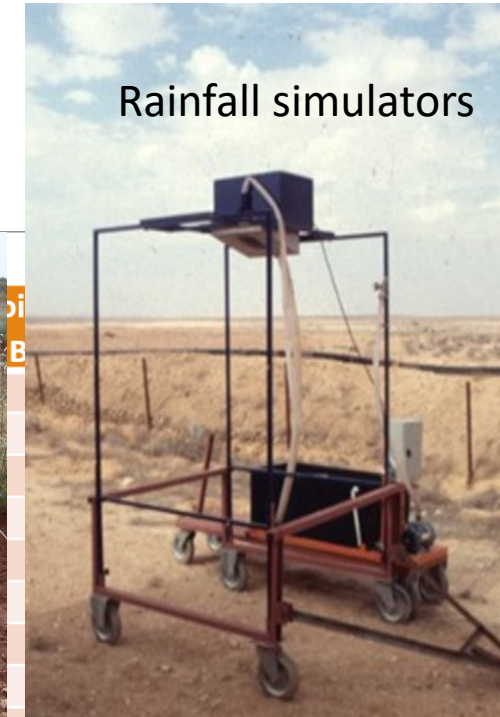
Soil Texture	Total soil water storage capacity (at Field Capacity)		Unavailable water for plants (at Wilting Point)		Available Water Holding Capacity	
	% by volume	mm/m of soil	% by volume	mm/m of soil	% by volume	mm/m of soil
Coarse sands	7.0	70	2.0	20	5.0	50
Fine sands	10.0	100	3.0	30	7.0	70
Loamy sand	15.0	150	5.0	50	10.0	100
Sandy loam	20.0	200	6.0	60	14.0	140
Loam	25.0	250	7.0	70	18.0	180
clay loam	30.0	300	10.0	100	20.0	200
clay	40.0	400	21.0	210	18.0	190

Step 6: determine design runoff coefficient (RC)

- Runoff Coefficient (RC%) = $\frac{\text{Volume of runoff}}{\text{Volume of rainfall}} \times 100$
- Runoff Coefficient (RC%) = $\frac{\text{Volume of runoff (m3)}}{\text{Rainfall (mm)/1000 x area (m2)}} \times 100$

RC value depends on:

- Land slope (the steeper the higher the RO)
- Rain intensity (the more intense rain the higher RC)
- Surface smoothness (the smoother the surface the higher RC)
- Soil Infiltration rate (The higher infiltration rate the lower RC)
- The length of the catchment (the longer the run the lower the RC)

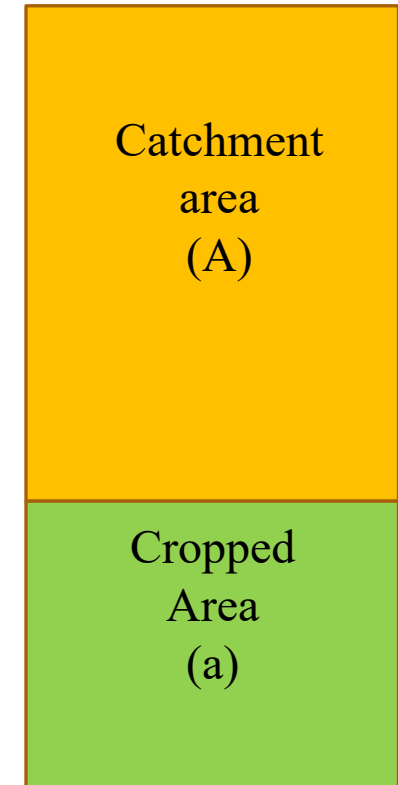


	≤0.5	11	16	25	26
Crops	0.5				28
	5-1				31
	≥1				35
	≤0.5				31
Bare soil	0.5				33
	5-1				36
	≥1				42
	Al				
Soil group A (Sand, loamy sand)					
Soil Group C (sandy clay loam)					
					and clay)

Step 7: Determining Catchment- Target ratio

ET = Evapotranspiration (annual)
R = rainfall (annual)
a = cultivated area
A = catchment area
Rc = Runoff coefficient

$$A/a = (ET-R)/(R * Rc)$$

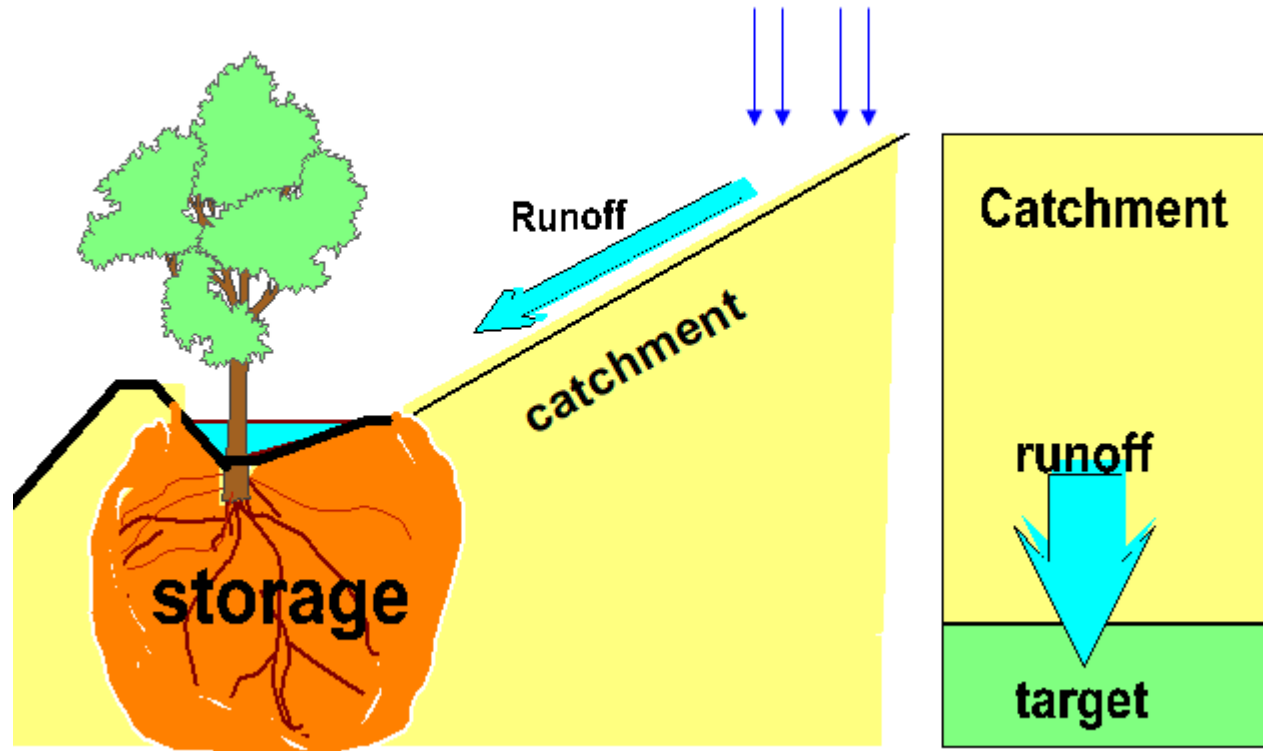


Step 8.

Check if the soil storage capacity can accommodate the runoff

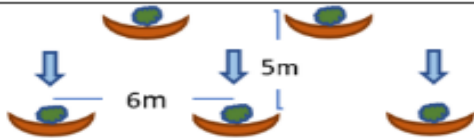
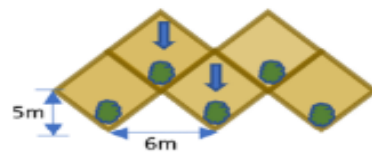
- Match the total volume of runoff plus direct rainfall to the total storage capacity of the soil
- Storage capacity is dynamic as water is taken from the soil by the crop during the growing season while runoff is continuing
- If storage is not enough, then we either change the crop to reduce water requirements or improve the soil water holding capacity which is normally costly (adding polymers etc.)

Microcatchments

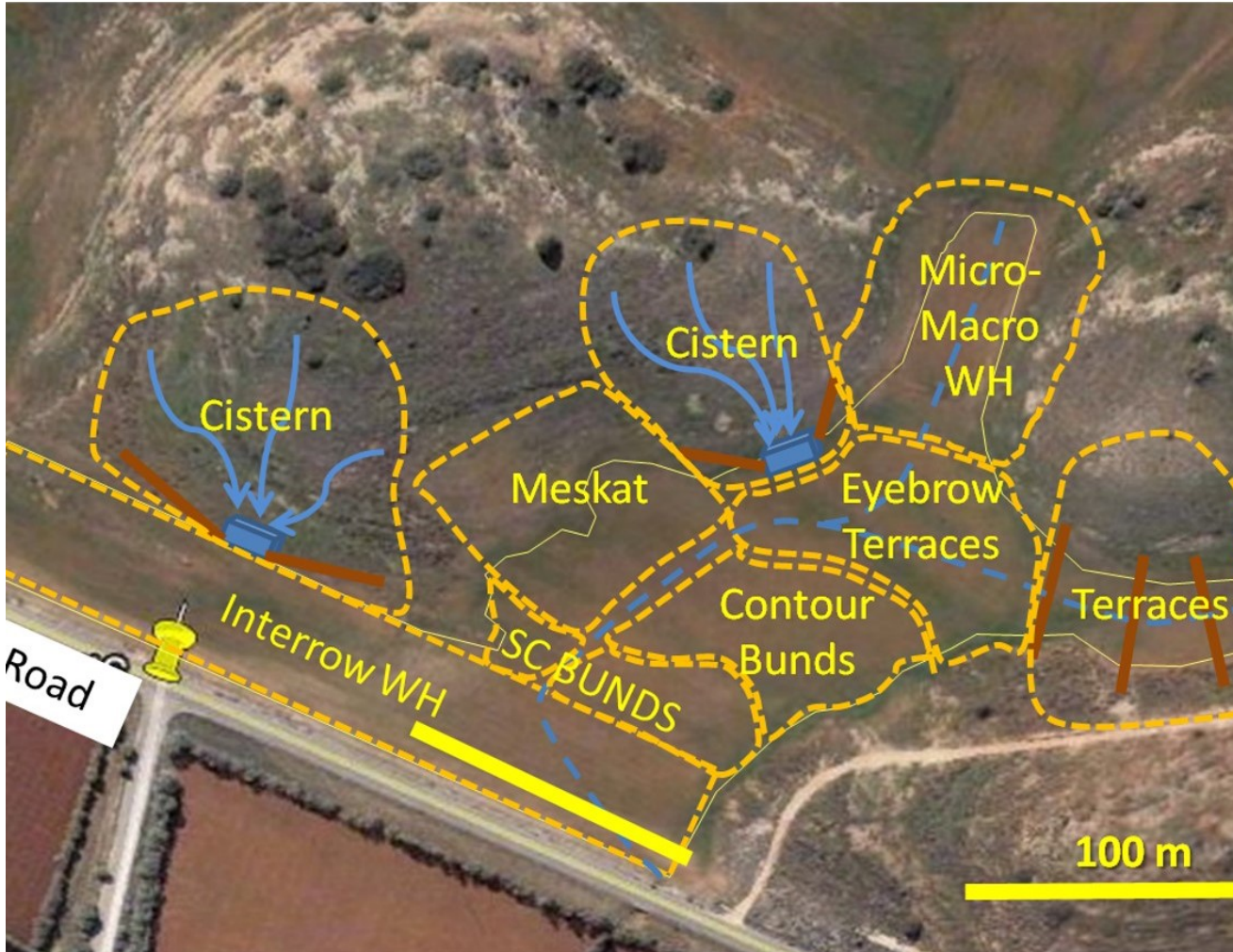


Design example for olive orchard

Table 9. Practical design example 2. Olive orchard with semicircular bunds & small basins

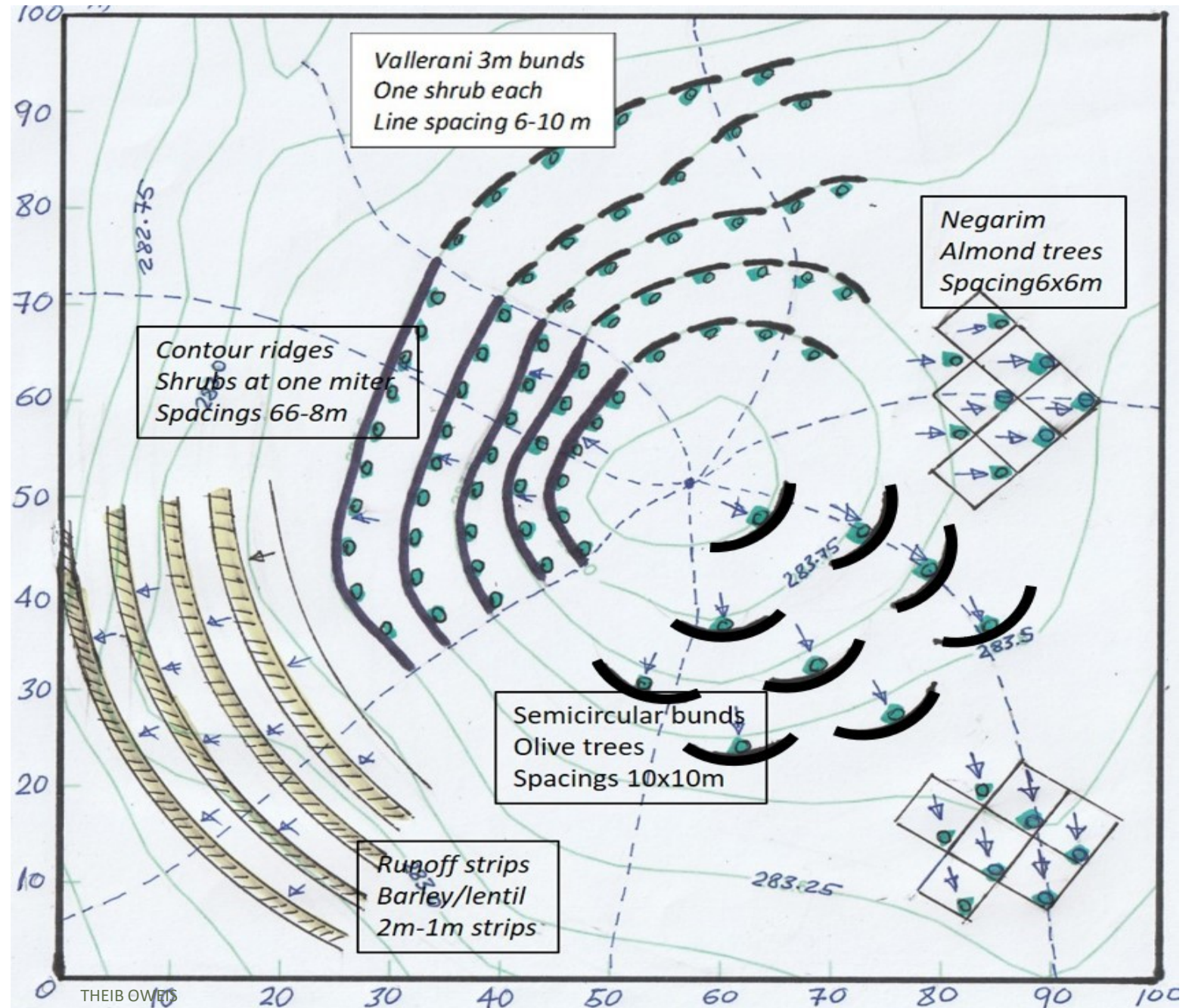
Step 1	Determine the need and potential for water harvesting	Olive trees in Jerash areas is grown in annual rainfall zone of 300-400 mm at times when olive trees need 500 mm annually. Trees survive the deficit but yields are low and fluctuate greatly from one year to the other. Availability of supplemental water through water harvesting will improve yields and provide more stable income and higher oil quality.
Step 2	Select the technique and the suitable site	A site in <u>Jarash</u> area with mean annual rainfall of 400 mm is selected on land with a slope of 5% having deep (1.5 m) clay loam soils. Consulting with table 1 suitable techniques are semicircular bunds and small basins (<u>negarim</u>)
Step 3	Determine crop water requirements and water deficit	From table 2. trees ET is around 500 mm. Rainfall of 400 mm the deficit would be 100 mm
Step 4	Determine design annual rainfall	Annual rainfall at 67% probability in the area was calculated by the equation to be 320 mm
Step 5	Determine design water storage capacity	- From table 5. A clay loam soil has 20% available water. - Olive matured trees is 3 m diameter (7m² area) and 1.5 m soil depth. - Available water per tree = 7.0m² * 1.5m * 20% = 2.1 m³
Step 6	Determine design runoff coefficient	For clay loam soil with slope of 5%, from table 7 Orchards similar to forest RC = 21% is suitable.
Step 7	Determine catchment /target ratio (A/a)	Applying the equation: $A/a = (ET-R)/(R*RC)$ - ET= 500 mm, Design R= 320, RC= 21% - $A/a = (500-320)/(320*0.21) = 2.7$, may be rounded to 3 then $A/a = 3:1$ - As the target area/tree is 7m² then the catchment area should be 21m² - Total area of catchment and target = 7+21=28 m² so best-rounded tree spacing = 6m x 5 m
Step 8	Check if the soil storage capacity can accommodate the expected runoff and rainfall	- Catchment area= Total tree area-Target area=(5m*6m)-(7.0 m²)= 23m² - Runoff volume= Catchment * rainfall * Runoff coefficient= 23.0m²*0.32m* .20= 1.47 m³ - Rainwater volume on target= Rainfall depth * area = 0.32 m*7m²= 2.24 m³ - Total water expected at target = Runoff volume + Rainfall volume = 1,47+ 2.24= 3.71 m³ - Storage capacity in soil profile= tree area*soil depth*available WHC = 7.0m²*1.5m*0.2=2.1m³. - Depleted ET volume from November to April Table 2 = ET (0.170m)*tree area (7.0m²)= 0.17m*7.0m²=1.19m³ - Total storage available=available WHC + depleted ET= 2.1+1.19 m³= 3.29 m³. This is a bit short of 3.71 m³ of water expected but within the error margin
Summary		Olive trees planted at spacings on 5x6 meters with semicircular bunds/small basins would provide the trees with supplemental water to cover the deficit between rainfall and crop ET. 

Using google maps for site selection and allocating water harvesting techniques



- On farm or at landscape scales
- Information on:
 - Topography
 - Crops
 - Rainfall storms amount and intensity
 - Soils physical properties
 - Runoff coefficients

Layout and mapping designed techniques





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Thank you for your attention



Boost your crop
water productivity
in selected farming
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